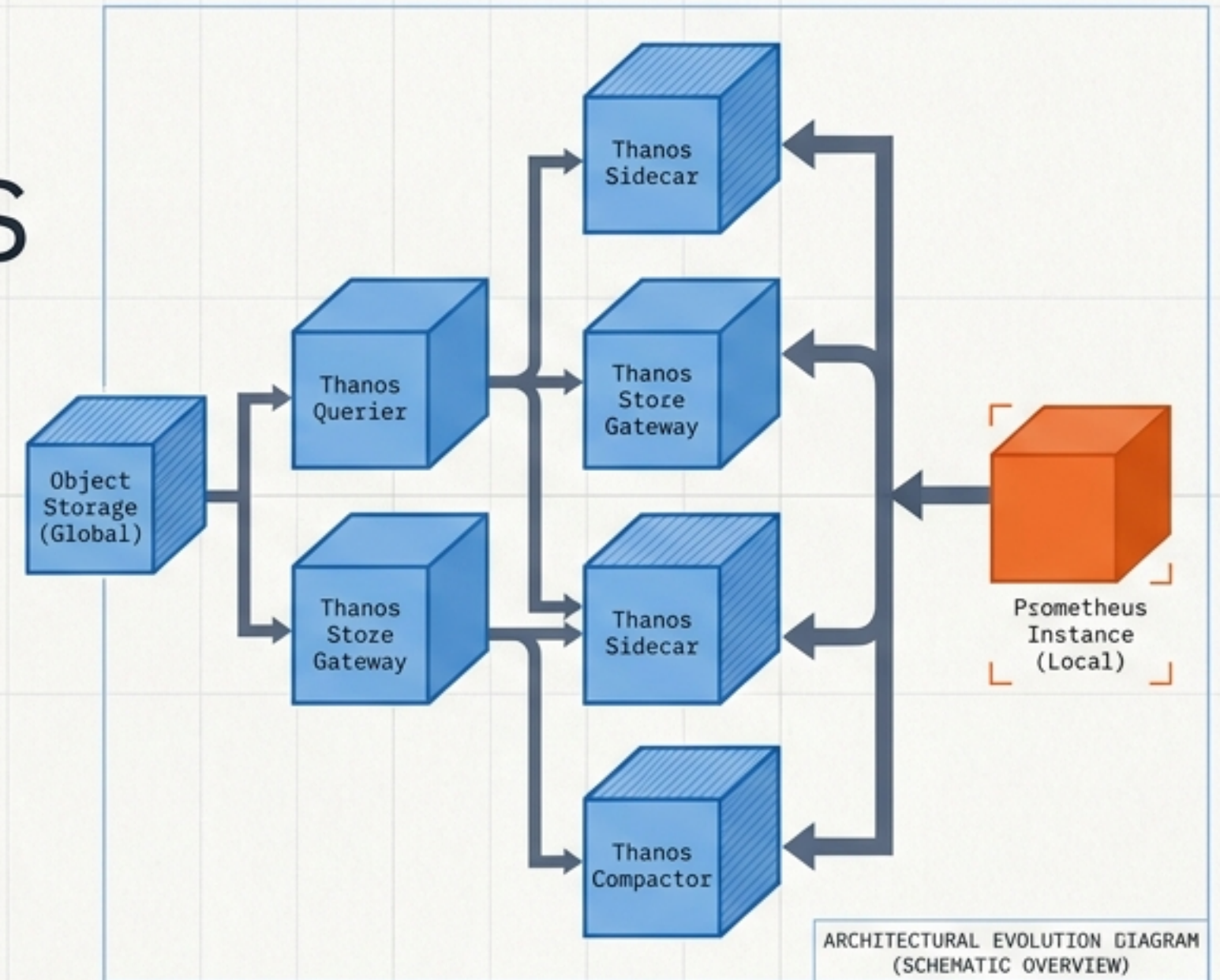
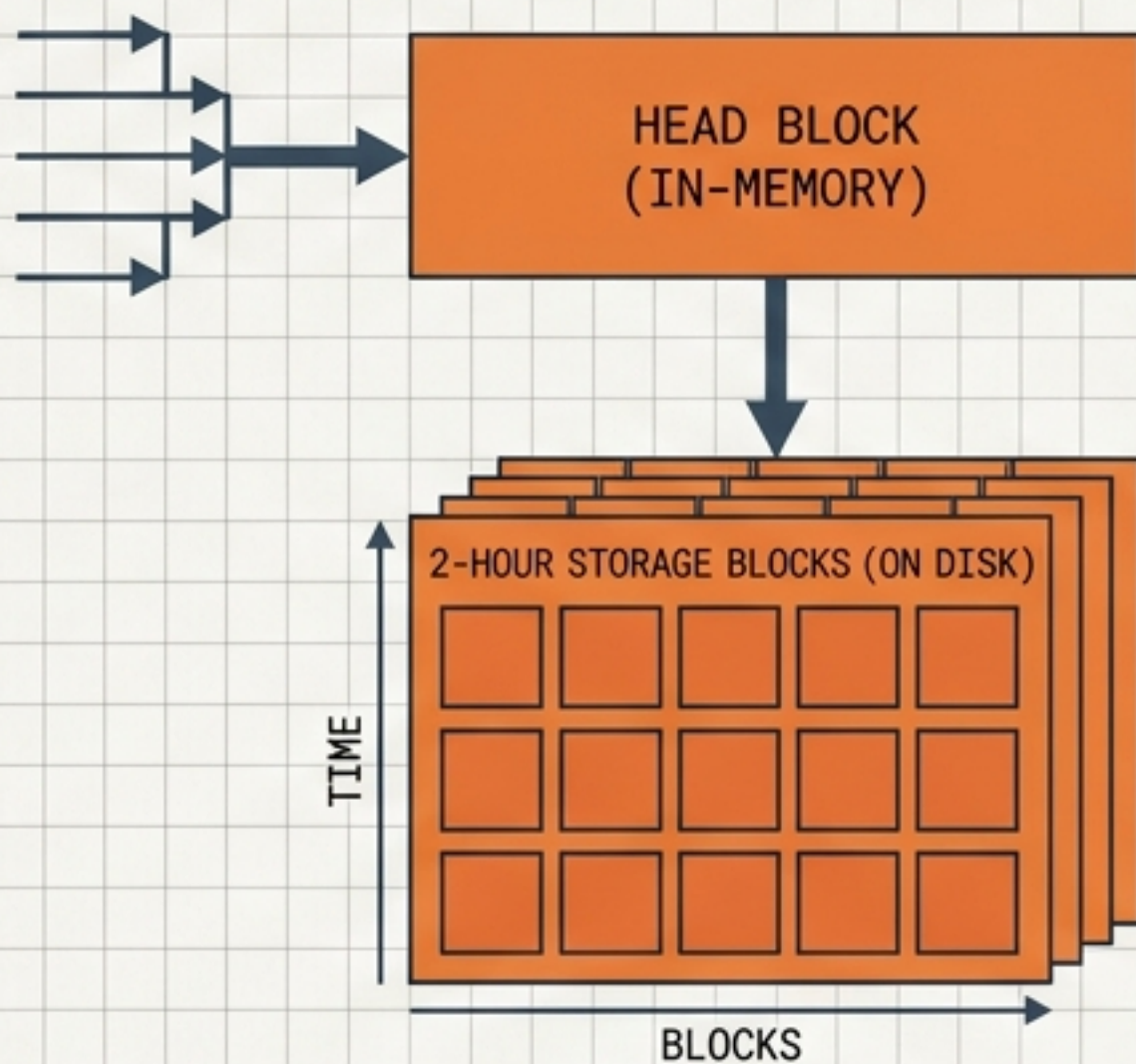


Scaling Prometheus with Thanos

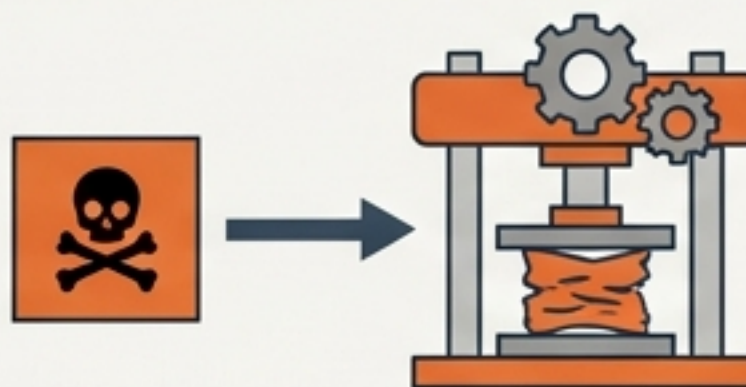
A definitive architectural playbook for preventing OOMs, avoiding race conditions, and decoupling long-term telemetry storage.



LOCAL PROMETHEUS NODE



THE TOMBSTONE MECHANIC



Data is marked for deletion (tombstone) but remains in the block until physically removed by the Compactor.

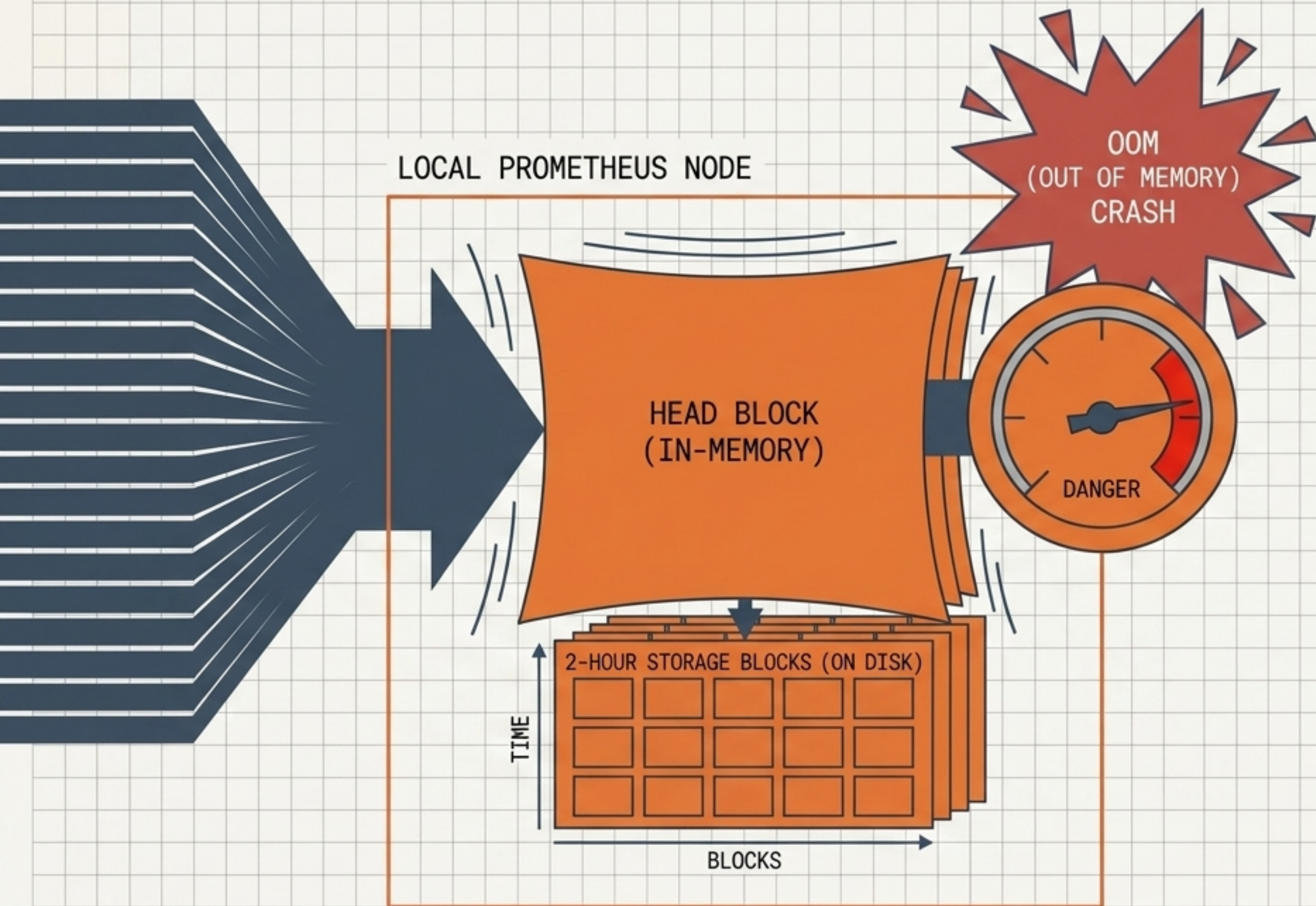
WRITE-AHEAD LOG (WAL)

MYTH

~~The WAL is a buffer holding data, waiting to write it to disk.~~

REALITY

The WAL exists purely for crash recovery. It is a safety net ensuring in-memory data is not lost if Prometheus restarts before a 2-hour block flush. ✓

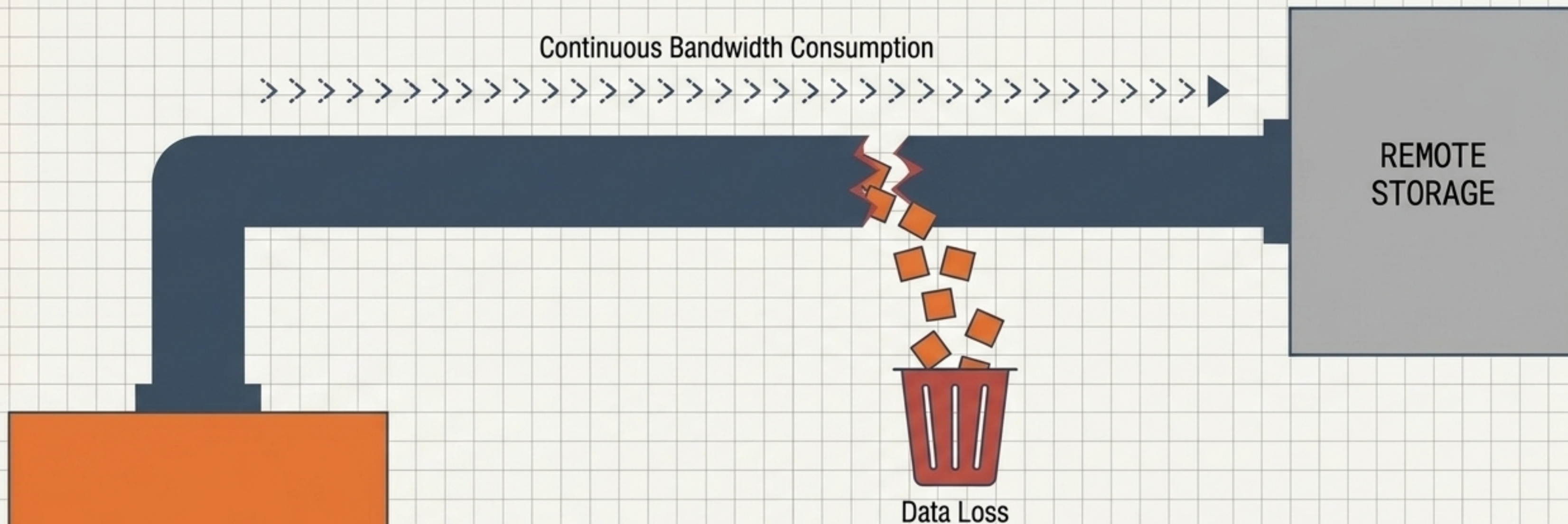


THE SINGLE-NODE LIMIT

Prometheus has no native clustering. It relies entirely on vertical scaling—requiring progressively larger CPU and RAM on a single instance.

THE BREAKING POINT

Heavy scrape volumes at a specific point in time cause the head block to rapidly expand, applying extreme memory pressure that leads to system slowdowns or total crashes.



TEMPORARY FIX: REMOTE WRITE

Mechanism:

Prometheus constantly pushes scraped samples to remote storage over the network.

The Trade-offs:

- **Cost:** Consumes heavy, continuous network bandwidth.
- **Vulnerability:** Network errors can cause unrecoverable data loss, mitigated—but not eliminated—by the remote-write retry queue.

THANOS GLOBAL QUERY LAYER

MYTH:

The Sidecar stores data for 2 hours before sending it.

FACT:

Prometheus buffers data locally for 2 hours. The Sidecar is just a lightweight process running next to it, watching for completed blocks.

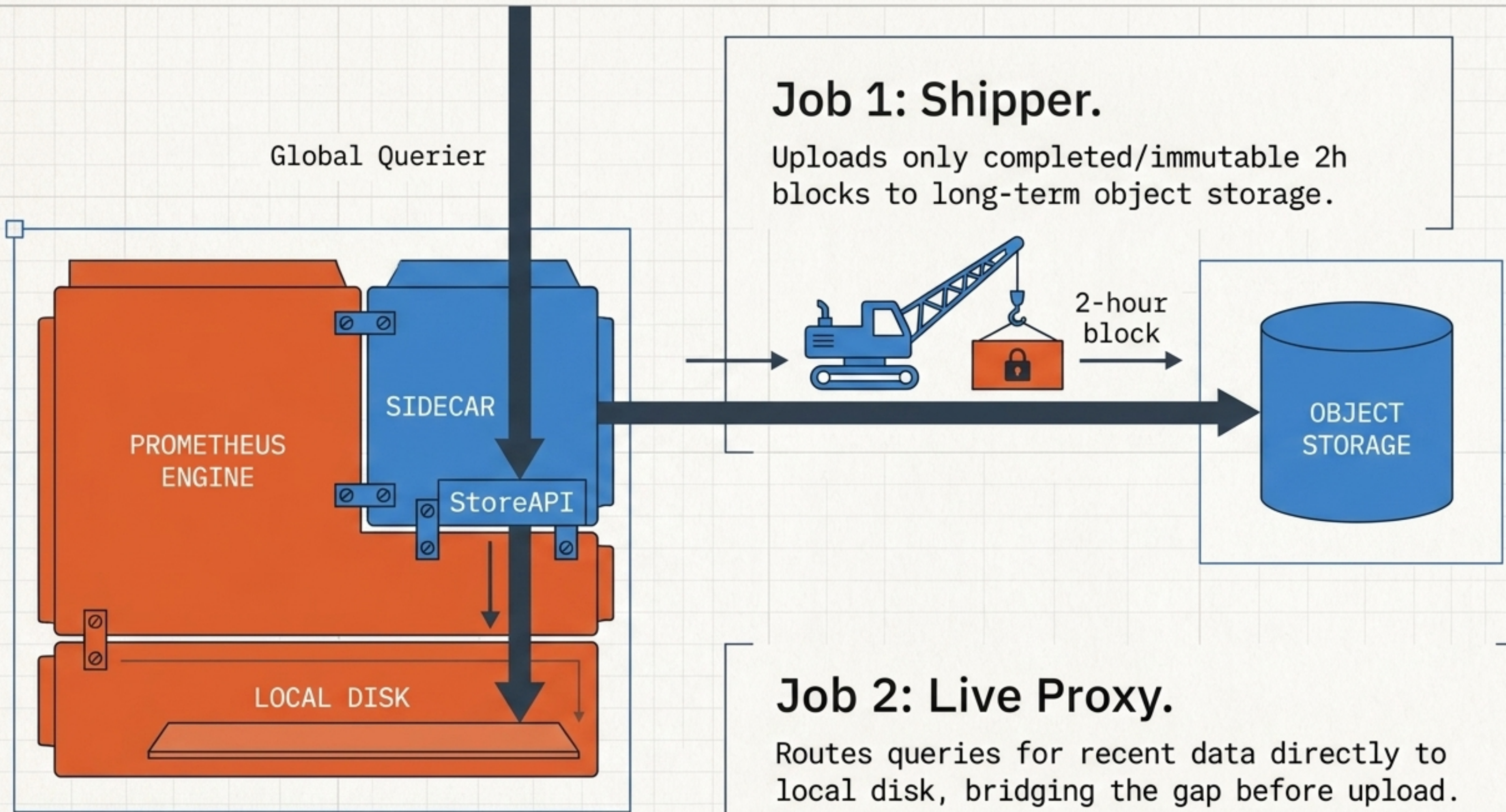
LOCAL PROMETHEUS NODE

HEAD BLOCK
(IN-MEMORY)

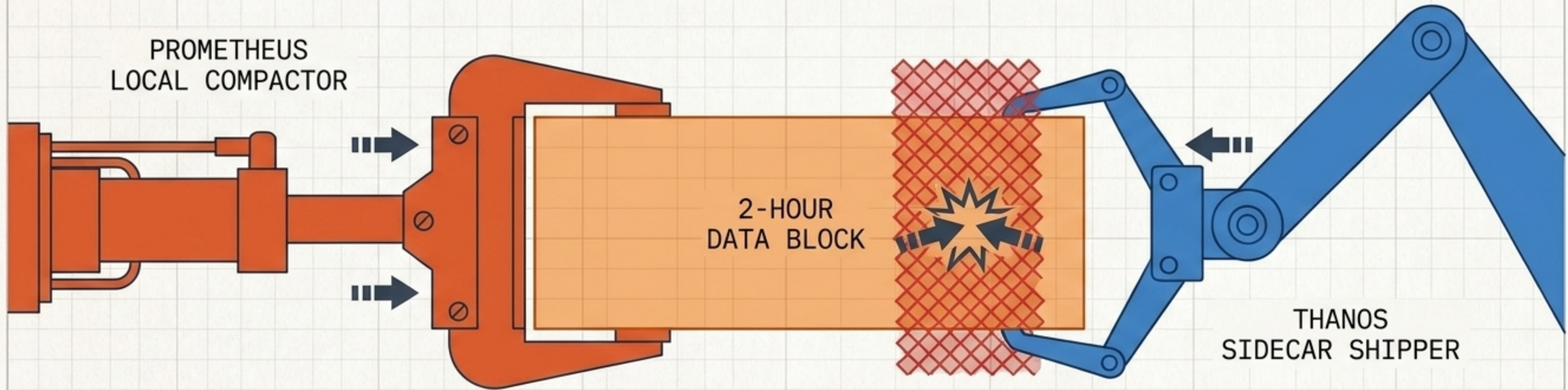
2-HOUR STORAGE BLOCKS
(ON DISK)

THANOS
SIDECAR

OBJECT STORAGE /
S3 BUCKET



THE COMPACTION COLLISION

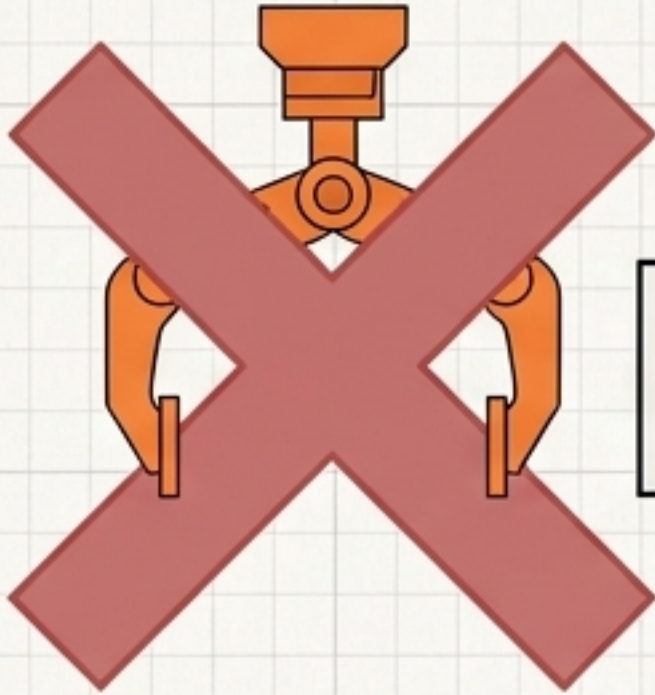


THE RACE CONDITION

If the local Prometheus compactor rewrites a block while the sidecar is uploading it (or just after), the local disk and object storage fall into an inconsistent, corrupted state.

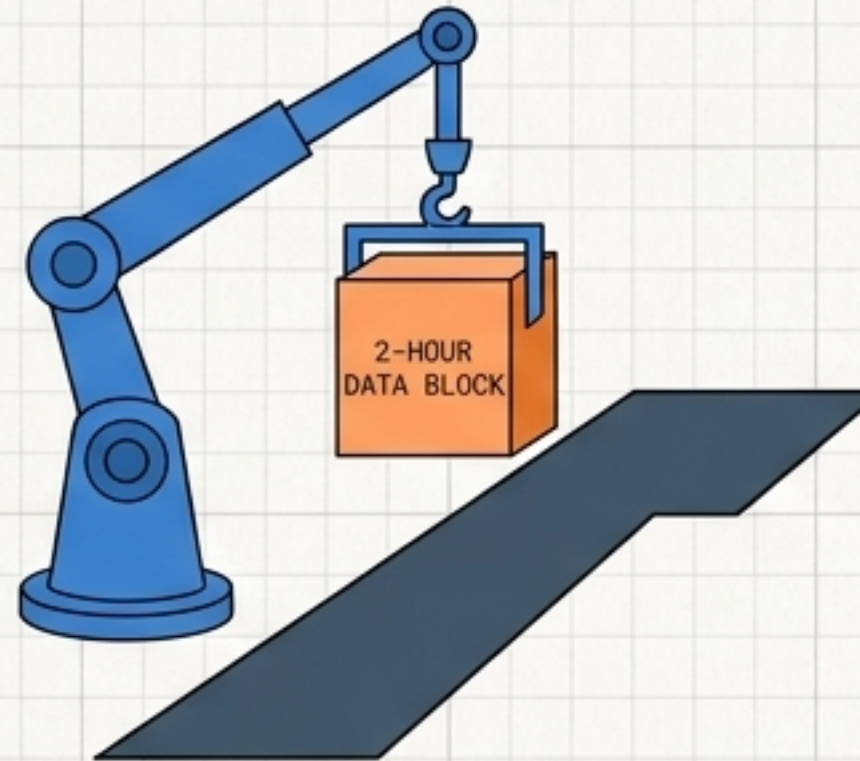
THE COMPACTION RESOLUTION

PROMETHEUS
LOCAL COMPACTOR

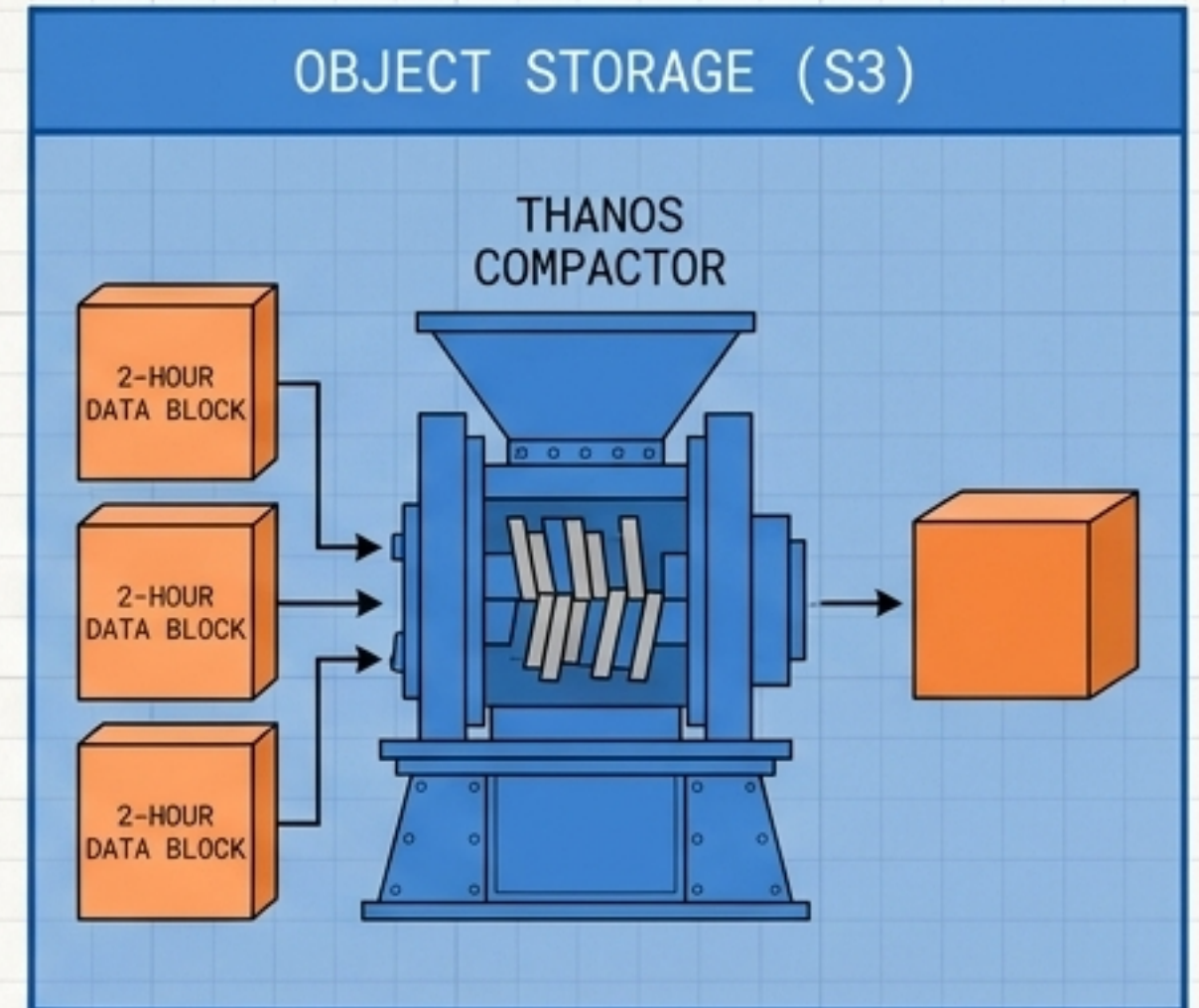


min = max
(Disable local
compaction)

THANOS
SIDECAR SHIPPER



OBJECT STORAGE (S3)

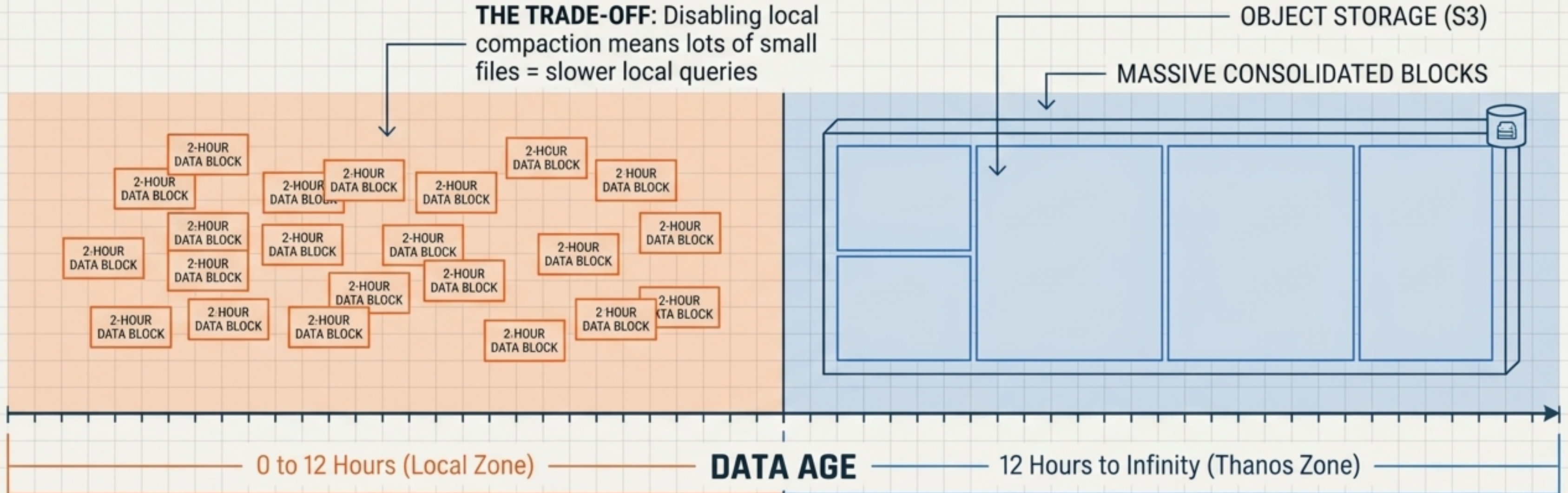


THE FIX:

Set **min = max** (typically 2h) to strictly disable local compaction. Every block stays small and ships as-is. Compaction is delegated downstream to the Thanos Compactor, operating safely against immutable data in S3.

THE COUNTERMEASURE: STRICT, SHORT LOCAL RETENTION

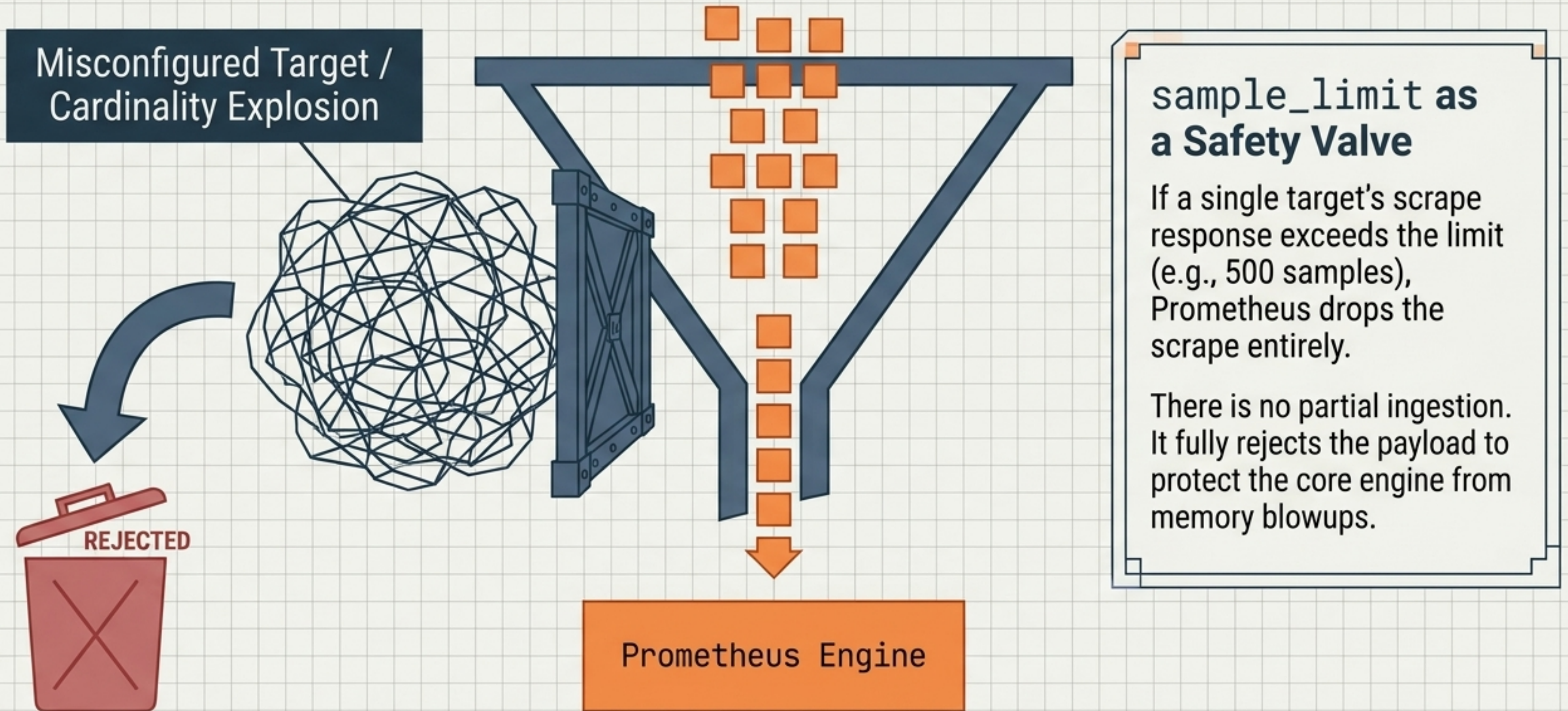
THE TRADE-OFF: Disabling local compaction means lots of small files = slower local queries



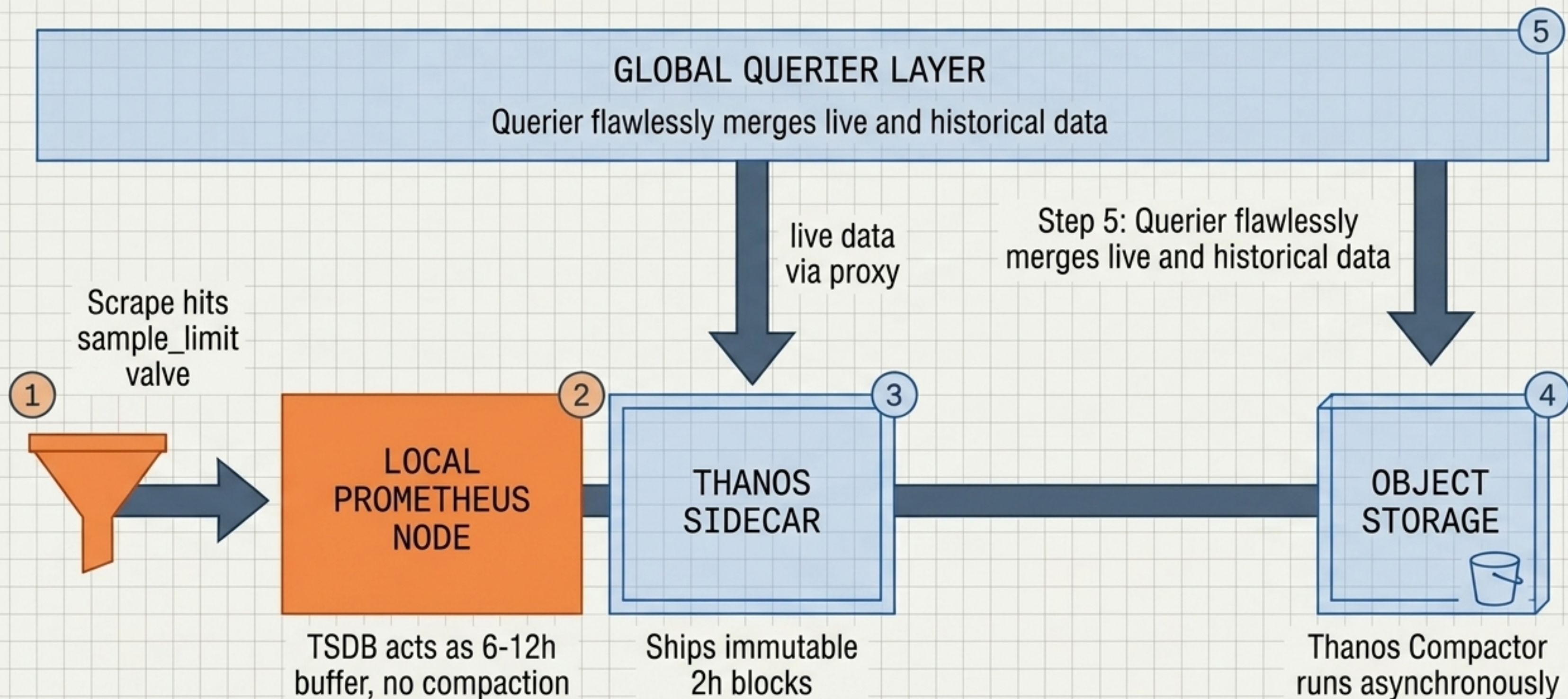
THE COUNTERMEASURE: STRICT, SHORT LOCAL RETENTION

Enforce a 6–12 hour retention limit on Prometheus. The local disk transforms from a long-term database into a lightweight, transient buffer. Older data queries are seamlessly handled by Thanos layers.

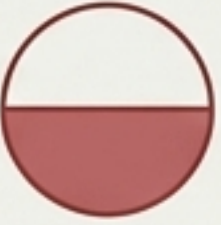
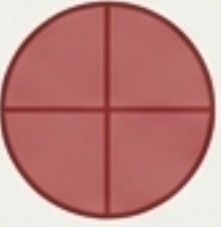
THE CARDINALITY SHIELD



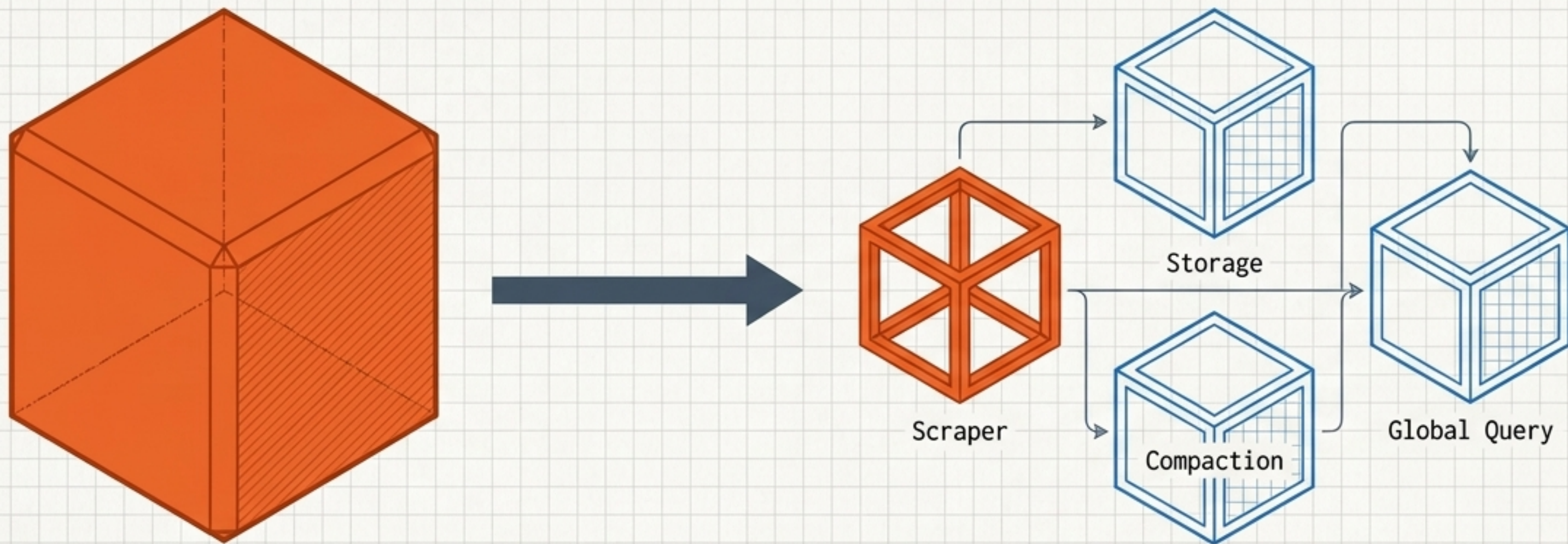
THE ULTIMATE SPATIAL BLUEPRINT: FULL ASSEMBLY



ARCHITECTURAL DECISION RECORD (ADR) MATRIX

Decision	Architectural Benefit	Operational Cost
Remote Write	Industrial Safety: Offloads storage from Prometheus.	Bandwidth heavy , potential data loss on network failure. 
Disable Local Compaction (min=max)	Architectural Blueprint: Eradicates Sidecar / Thanos. / Compactor race conditions.	Slower local queries due to unoptimized block fragmentation. 
Short Local Retention (6-12h)	Industrial Safety: Offsets slow local queries; keeps local disk highly performant.	Forces reliance on S3/Thanos for data > 12h, adding marginal query latency. 
Configure sample_limit	Industrial Safety: Shields against Promeees catastrophic cardinality blowups.	Misbehaving targets lose 100% of their scrape data until fixed. 

The Ultimate Goal: Decoupling Storage from Computation



Before: Prometheus acts as a heavy, stateful monolith attempting to scrape, store, compress, and query simultaneously.

After: Prometheus is transformed into a lightweight, transient scraper. All heavy lifting is delegated to specialized, stateless Thanos microservices. Infinite horizontal scale. Zero race conditions. Total reliability.